

1. Administrative & Study Identification

Full Study Title: A Study to Learn About a Combined COVID-19 and Influenza Shot in Healthy Adults **Short Title/Acronym:** Combined COVID-19/Flu Vaccine Study **Protocol Number:** C5681001 **NCT Number:** NCT06237049 **Posting ID:** 439709764418622968 **Sponsor Name:** Pfizer **Company:** Pfizer **Investigational Product:** Licensed BNT162b2 (Omi XBB. 1.5) and recombinant influenza vaccine (RIV) administered together as a single injection (BNT162b2 [Omi XBB. 1.5]/RIV) **Preferred UMLS Name:** COVID-19 and Influenza Virus Vaccine **Phase of Development:** PHASE2 (Phase 1/2) **Trial Status:** COMPLETED **Medical Monitor:** Pfizer Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the safety, tolerability, and immunogenicity of licensed BNT162b2 (Omi XBB. 1.5) and recombinant influenza vaccine (RIV) administered together as a single injection in healthy adults 50 years of age or older. **Secondary Objectives:** To compare the safety, tolerability, and immunogenicity of the single injection to BNT162b2 (Omi XBB. 1.5) and RIV administered simultaneously as 2 separate injections (coadministered), and to BNT162b2 or RIV when administered alone.

2.2 Background & Rationale To see if combining a licensed COVID-19 vaccine and a licensed influenza vaccine into a single shot is safe and can help produce antibodies to defend the body against both SARS-CoV-2 (the virus that causes COVID-19) and influenza. The potential benefit of a combination vaccine is improved vaccine uptake by requiring fewer injections and reducing the administration burden.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Active Comparator (separate injections of BNT162b2 and RIV) and Placebo (Normal saline placebo) **Blinding:** Observer-blind/Double-blind **Randomization:** Randomized into different parallel groups (Group 1, 2, 3, and 4)

3.2 Planned Enrollment & Duration Targeted Enrollment: 644 participants **Number of Sites:** Multicenter **Estimated Study Start:** 30-Jan-2024 **Estimated Primary Completion:** Completed

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Male or female participants aged 50 years or older at Visit 1 (Day 1). 2. Willing and able to comply with all scheduled visits, vaccination plan, laboratory tests, lifestyle considerations, and other study procedures. 3. Capable of giving signed informed consent as described in the protocol.

4.2 Exclusion Criteria 1. Immunocompromised individuals with known or suspected immunodeficiency. 2. Current heart disease, uncontrolled hypertension, or a prior history of myocarditis or pericarditis. 3. Prior history of ischemic stroke, transient ischemic attack, or Guillain-Barré syndrome (GBS). 4. Vaccination with any investigational or licensed influenza or COVID-19 vaccine within 6 months (180 days) before study intervention administration. 5. Women who are pregnant, plan to become pregnant during the study, or are breastfeeding. 6. Participants with a calculated BMI of ≥ 35 . 7. Known infection with human immunodeficiency virus (HIV), hepatitis C virus (HCV), or hepatitis B virus (HBV). 8. Any medical or psychiatric condition including recent (within the past year) or active suicidal ideation/behavior. 9. History of autoimmune disease requiring therapeutic intervention.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Single combined injection of BNT162b2 (Omi XBB. 1.5) and RIV. **Route of Administration:** Intramuscular injection. **Duration of Treatment:** Single administration (Day 1).

5.2 Concomitant & Prohibited Therapy Permitted: Standard daily medications that do not possess immunosuppressive properties. **Prohibited:** Receipt of chronic medications with known systemic immunosuppressant effects (including cytotoxic agents or systemic corticosteroids, e.g., triamcinolone), radiotherapy within 60 days before enrollment, or receipt of blood/plasma products, immunoglobulin, or monoclonal antibodies used for the treatment or prevention of COVID-19.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures	:---
:---	:---	Screening	Day -28 to 0 Informed Consent, Medical History, Baseline Labs
Baseline	Day 1	Randomization, Vaccination, e-diary provision	Interim 1
Week 4	Immunogenicity blood draw, safety assessment, e-diary review	End of Study	Month 6 Final safety assessment, SAE monitoring, Exit Interview

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Immunogenicity measured by Geometric Mean Titers (GMTs) of SARS-CoV-2 Omicron XBB.1.5 and influenza strains (A/Victoria/4897/2022 [H1N1], A/Darwin/9/2021 [H3N2], B/Michigan/1/2021, and B/Phuket/3073/2013) at 4 weeks after vaccination. Seroresponse rate defined as $\geq 4LLQ$. *Measurement Method:** Serological Laboratory Analysis (HAI titers $\geq 1:40$).

7.2 Secondary & Exploratory Endpoints Percentage of participants reporting local reactions (redness, swelling, pain) and systemic events (fever, fatigue, headache, vomiting, diarrhea, chills, new/worsened muscle pain, new/worsened joint pain). Evaluation of local and systemic event severity (mild, moderate, severe, potentially life-threatening). Percentage of participants who reported any Adverse Events (AEs) from vaccination through 4 weeks.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Recorded by participants in an electronic diary (e-diary) following vaccination. **Serious Adverse Events (SAEs):** Tracked and reported over the study duration; defined as any untoward medical occurrence in a participant temporally associated with the use of study intervention. **Data Monitoring Committee (DMC):** Internal/External safety monitoring boards per Pfizer standard protocol.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Seroresponse evaluated for groups based on exact received regimens. **Sample Size Justification:** Sufficient to establish 95% Confidence Intervals for immunogenicity non-inferiority comparisons (Target enrollment of 644 randomized 1:1:1:1). **Handling of Missing Data:** Exact 2-sided Confidence Intervals (CI) calculated based on the Clopper and Pearson method for binary endpoints. Descriptive statistics utilized for continuous variables.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Scheduled site monitoring and continuous e-diary compliance checks. **Audit Trail:** Centralized tracking of e-diaries and laboratory sample chains of custody.

11. Study Identifiers & Governance

Primary ID: {{C5681001}} **Registry Number:** {{NCT06237049}} **Posting ID:** {{439709764418622968}} **Sponsor/Lead Organization:** {{Pfizer}} **Company:** {{Pfizer}} **Study Title:** {{A Study to Learn About a Combined COVID-19 and Influenza Shot in Healthy Adults}} **Official Regulatory Title:** {{A Phase 1/2 Randomized Study to Evaluate the Safety, Tolerability, and Immunogenicity of a Modified RNA COVID-19 Vaccine and a Recombinant Influenza Vaccine Administered as a Single Injection in Healthy Adults 50 Years of Age or Older}}

12. Clinical Framework & Terminology Mapping

Disease Areas: {{Influenza, Human, SARS-CoV-2 Infection, COVID-19}} **Condition Name:** {{Influenza}} **Condition Preferred Name:** {{Influenza}} **Condition Semantic Type:** {{Disease or Syndrome}} **Condition Definition:** {{An acute viral infection in humans involving the respiratory tract. It is marked by inflammation of the NASAL MUCOSA; the PHARYNX; and conjunctiva, and by headache and severe, often generalized, myalgia.}} **MeSH Code:** {{D007251}} **SNOMED ID:** {{6142004}} **Diagnosis Threshold:** {{Healthy Adults \$\ge\$ 50 years}} **Medical Methodology:** {{Prophylactic Vaccination}} **Optimization Target:** {{Simultaneous robust immunogenicity against SARS-CoV-2 and Influenza A/B}}

12.1 Drug Ontology Mapping (*Recorded to standardize prohibited systemic immunosuppressant tracking*) **Drug Name:** {{triamcinolone}} **Drug Preferred Name:** {{triamcinolone}} **ATC Code:** {{D07XB02}}

13. Study Procedures & Methodology

Management Pathway: {{Standard prophylactic vaccination administration}} **Education Protocol (HCP):** {{Investigational product handling and injection site assessment training}} **Education Protocol (Patient):** {{Onsite training for the accurate use of the electronic diary (e-diary) for safety reporting}} **Quality Audit Mechanism:** {{Patient e-diary completion tracking & laboratory protocol adherence}}

14. Observation & Follow-Up Schedule

Total Duration: {{Up to 6 months of safety follow-up}} **Onsite Visit Cadence:** {{Day 1, Week 4 (1 Month)}} **Remote Monitoring Frequency:** {{Daily tracking via e-diary for the first 7 days post-vaccination}} **Checklist Review Interval:** {{Ongoing automated e-diary review and standard AE collection periods}}

15. Sign-off & Approval

Role	Name	Signature	Date	:---	:---	:---	:---	
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]					
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]			Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]					